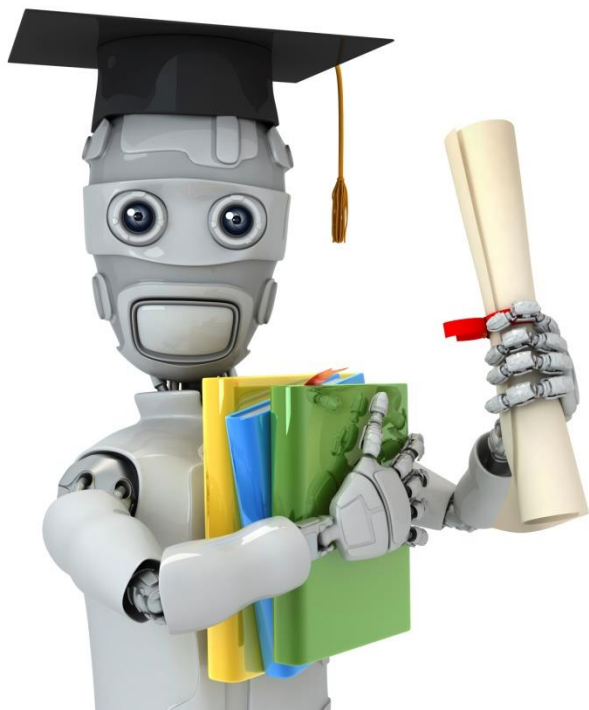




Machine Learning

# Logistic Regression

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# Classification

# Classification

Email: Spam / Not Spam?

Online Transactions: Fraudulent (Yes / No)?

Tumor: Malignant / Benign ?

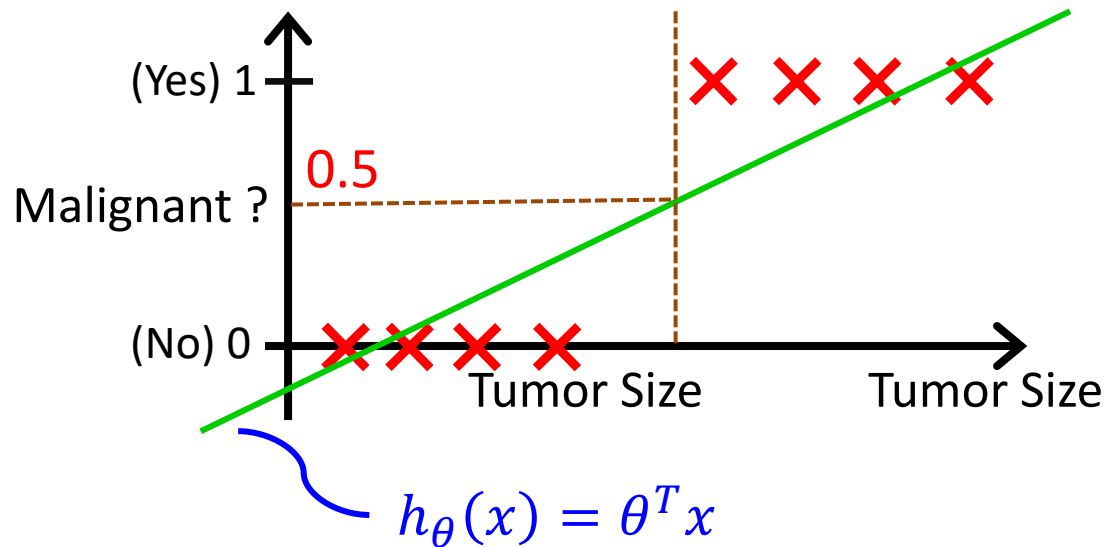
$$y \in \{0, 1\}$$


0: “Negative Class” (e.g., benign tumor)

1: “Positive Class” (e.g., malignant tumor)

**Binary classification:** classification problem with just two categories.

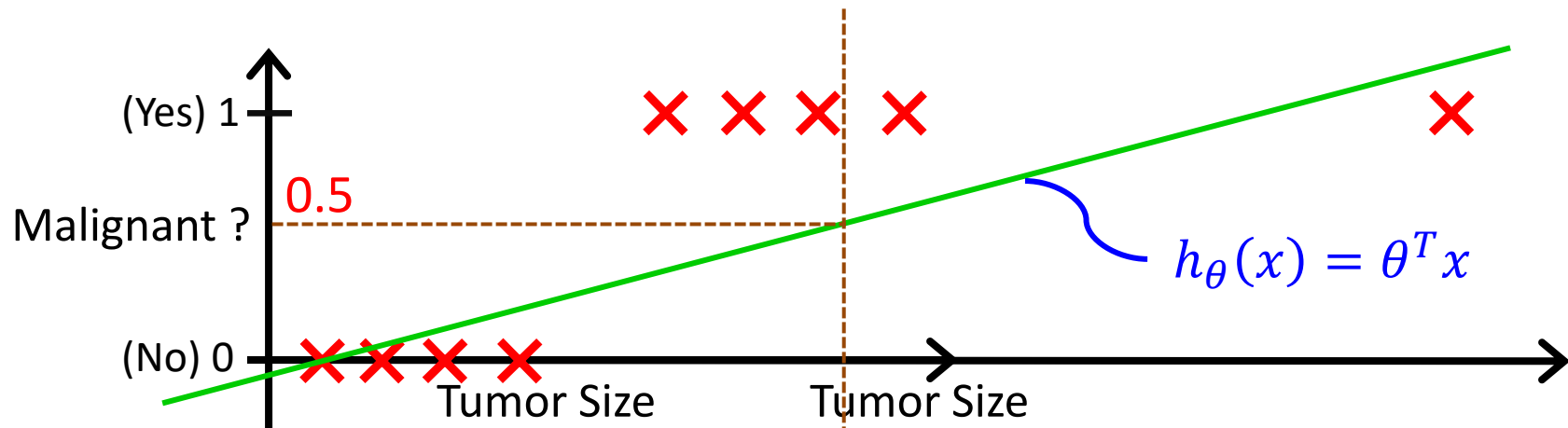
**Multi-class classification:** classification problem with more than two categories.



Threshold classifier output  $h_{\theta}(x)$  at 0.5:

If  $h_{\theta}(x) \geq 0.5$ , predict “y = 1”

If  $h_{\theta}(x) < 0.5$ , predict “y = 0”



Threshold classifier output  $h_{\theta}(x)$  at 0.5:

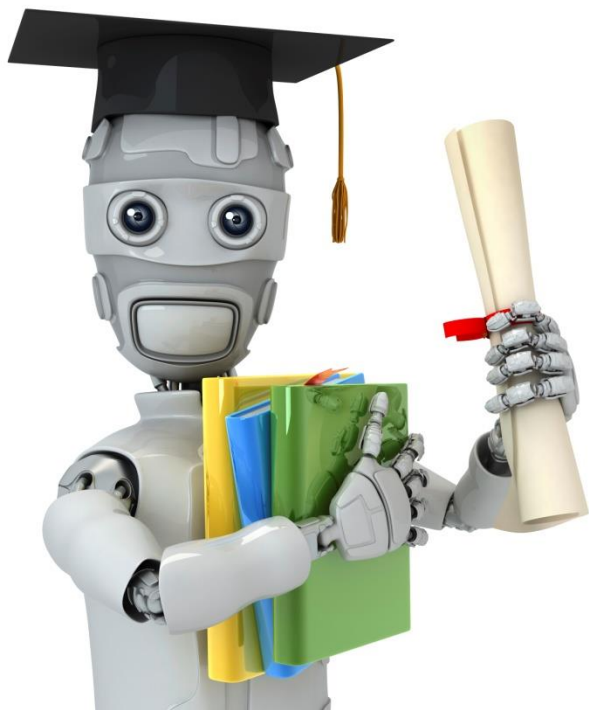
If  $h_{\theta}(x) \geq 0.5$ , predict “y = 1”

If  $h_{\theta}(x) < 0.5$ , predict “y = 0”

Classification:  $y = 0$  or  $1$

$h_{\theta}(x)$  can be  $> 1$  or  $< 0$

Logistic Regression:  $0 \leq h_{\theta}(x) \leq 1$



Machine Learning

# Logistic Regression

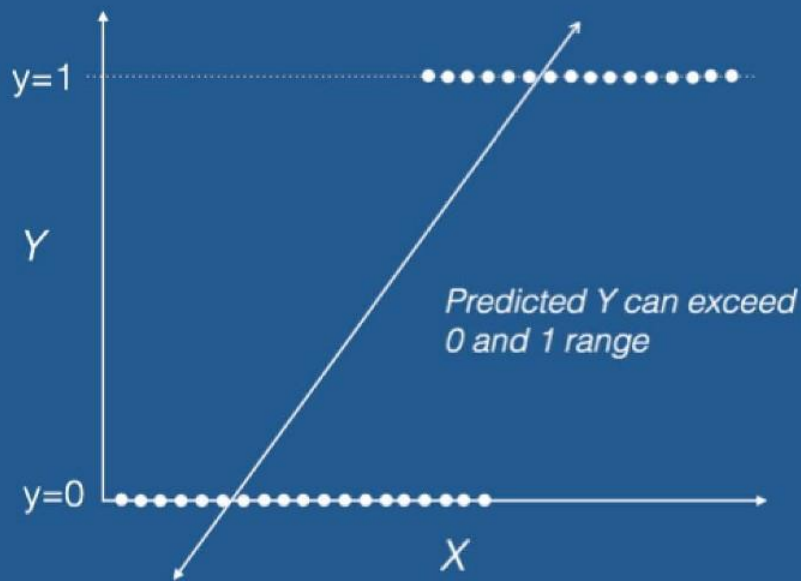
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# Hypothesis Representation

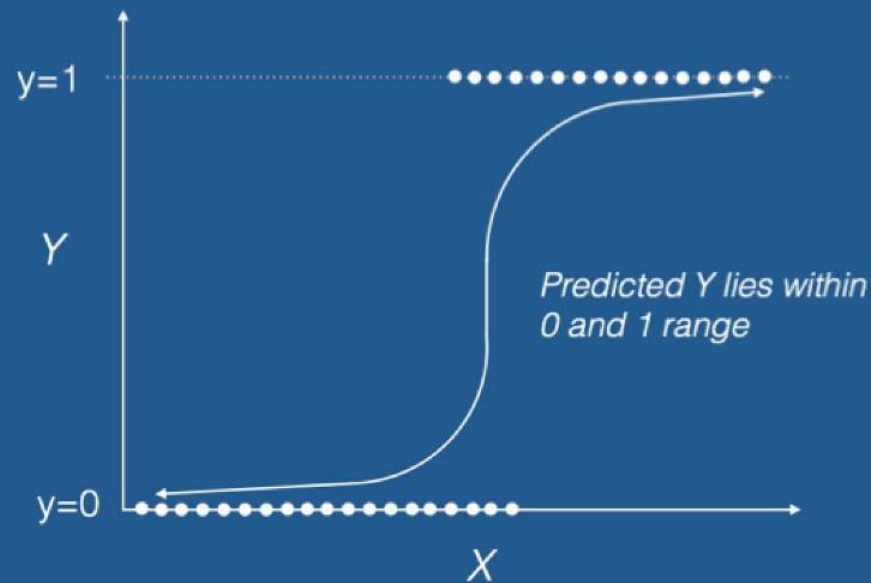
# Logistic Regression Model

Want  $0 \leq h_{\theta}(x) \leq 1$

## Linear Regression

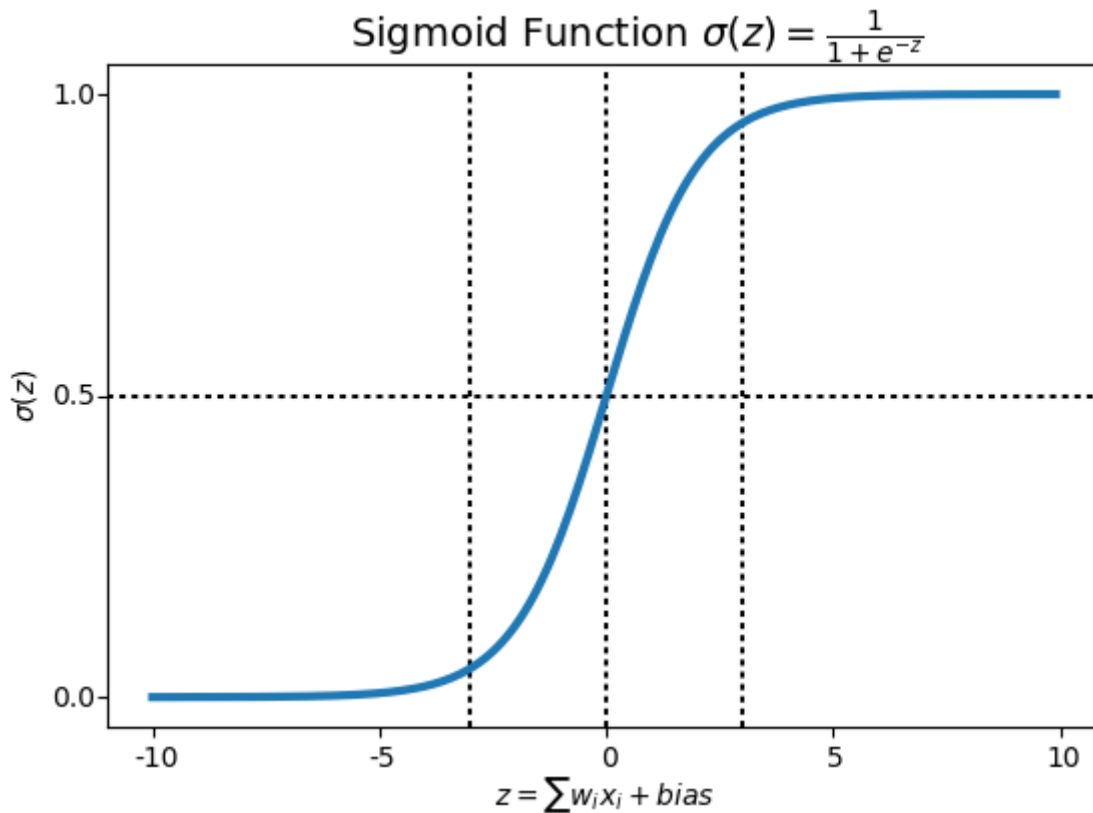


## Logistic Regression



# Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$





# Logistic Regression Model

Want  $0 \leq h_{\theta}(x) \leq 1$

Linear regression hypothesis:  $h_{\theta}(x) = \theta^T x$



Logistic regression hypothesis:  $h_{\theta}(x) = \underline{g(\theta^T x)}$


$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Sigmoid function

$$g(z) = \frac{1}{1 + e^{-z}}$$

## Interpretation of Hypothesis Output

$h_{\theta}(x)$  = estimated probability that  $y = 1$  on input  $x$

Example: If  $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$

$$h_{\theta}(x) = 0.7$$

Tell patient that 70% chance of tumor being malignant

Formally,

$$h_{\theta}(x) = P(y = 1|x; \theta)$$

“probability that  $y = 1$ , given  $x$ ,  
parameterized by  $\theta$ ”

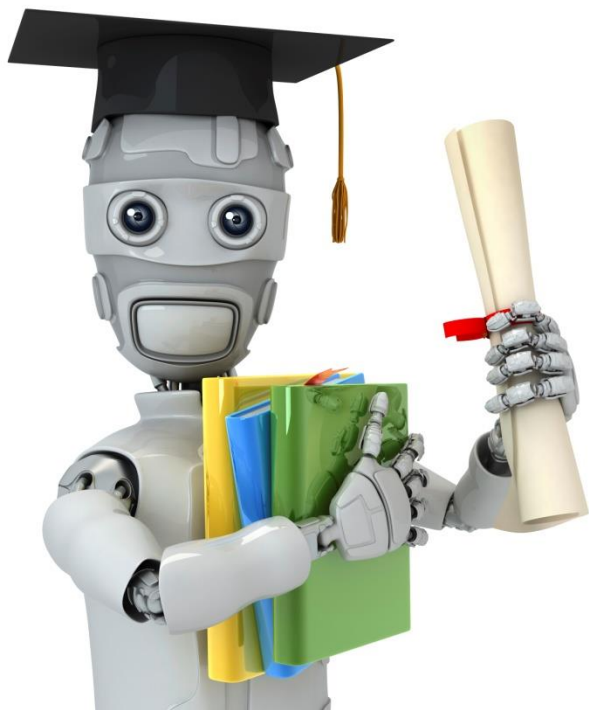
$$P(y = 0|x; \theta) + P(y = 1|x; \theta) = 1$$

$$P(y = 0|x; \theta) = 1 - P(y = 1|x; \theta)$$

# Logistic Regression Hypothesis

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

*The value of hypothesis is interpreted as the probability that the input  $x$  belongs to class  $y = 1$ . i.e. probability that  $y = 1$ , given  $x$ , parametrized by  $\theta$ .*



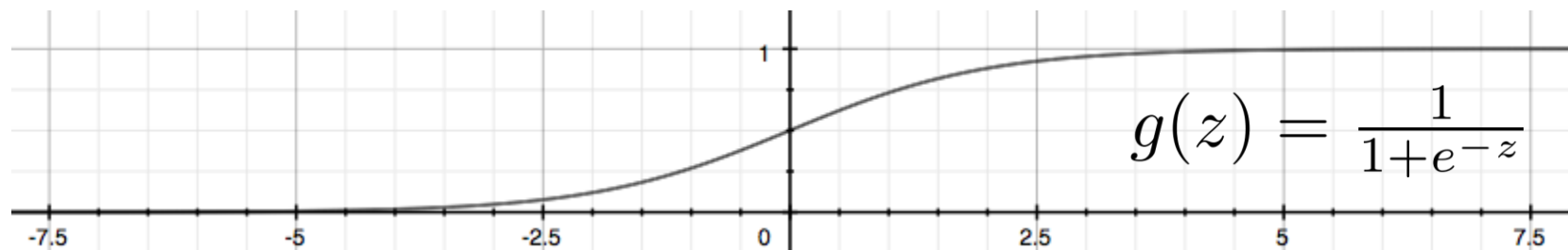
Machine Learning

# Logistic Regression

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## Decision boundary

**Logistic regression**      $h_{\theta}(x) = g(\theta^T x)$



**Suppose:**

Predict “ $y = 1$ ” if  $h_{\theta}(x) \geq 0.5$

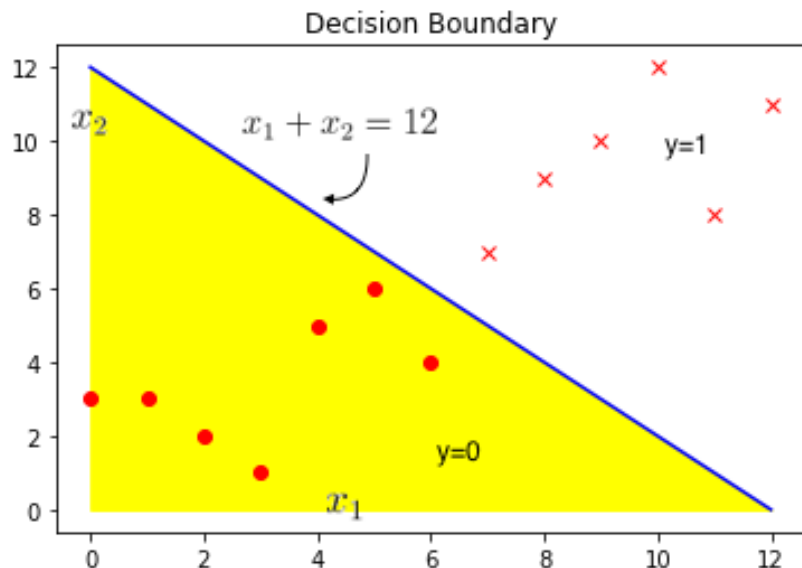
Predict “ $y = 1$ ” if  $\theta^T x \geq 0$

Predict “ $y = 0$ ” if  $h_{\theta}(x) < 0.5$

Predict “ $y = 0$ ” if  $\theta^T x < 0$

$g(z) \geq 0.5, \text{ if } z \geq 0$   
 $g(z) < 0.5, \text{ if } z < 0$

# Decision Boundary



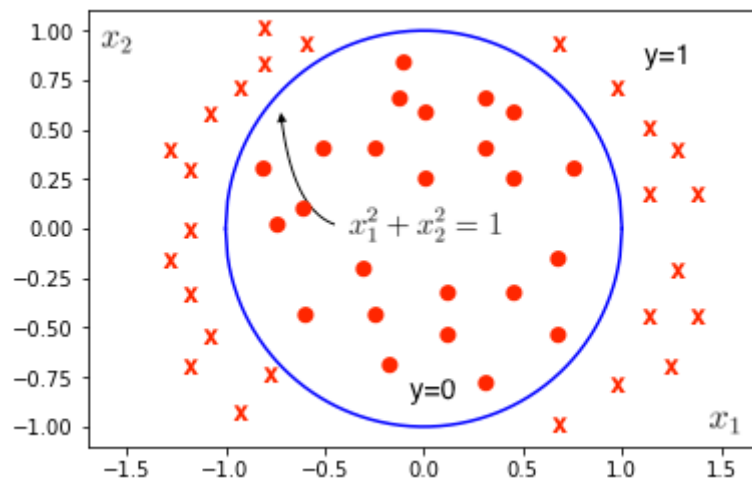
$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

$$\theta = [-12, 1, 1]^T$$

Predict  $y = 1$ , if  $-12 + x_1 + x_2 \geq 0$  or  $x_1 + x_2 \geq 12$

Predict  $y = 0$ , if  $-12 + x_1 + x_2 < 0$  or  $x_1 + x_2 < 12$

# Non-linear Decision Boundary



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

$$\theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

***As the order of features is increased, more and more complex decision boundaries can be achieved by logistic regression.***



Substituting (12) in (11),

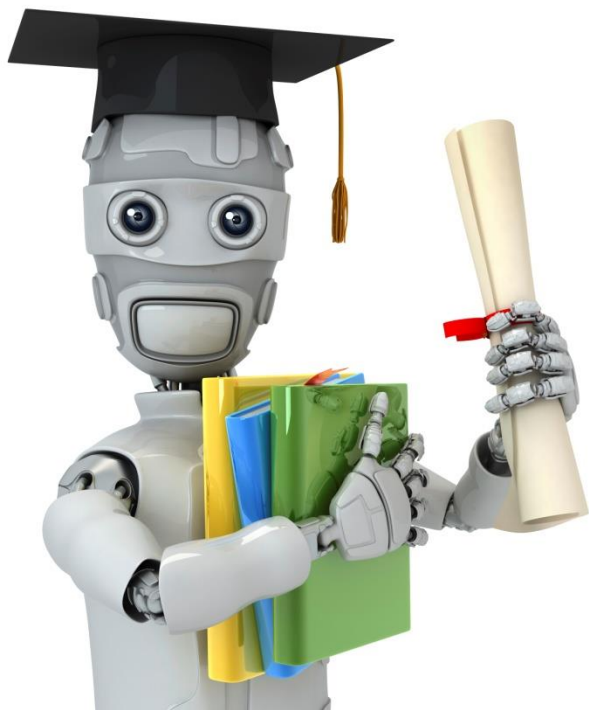
$$\theta^T x = -1 + x_1^2 + x_2^2$$

So, from (7), the decision boundary is given by,

$$\begin{aligned} -1 + x_1^2 + x_2^2 &= 0 \\ x_1^2 + x_2^2 &= 1 \end{aligned}$$

Which the equation of a circle at origin with radius 1, as can be seen in the plot above. And, using the  $\theta$  from (12) and hypothesis from (11), (7) can be written as,

$$\begin{aligned} \text{predict } y = 1, & \text{ if } -1 + x_1^2 + x_2^2 \geq 0 \text{ or } x_1^2 + x_2^2 \geq 1 \\ \text{predict } y = 0, & \text{ if } -1 + x_1^2 + x_2^2 < 0 \text{ or } x_1^2 + x_2^2 < 1 \end{aligned} \tag{12}$$



Machine Learning

# Logistic Regression

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## Cost function

Training set:  $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$

$m$  examples  $x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix} \quad x_0 = 1, y \in \{0, 1\}$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

How to choose parameters  $\theta$  ?

## Cost function

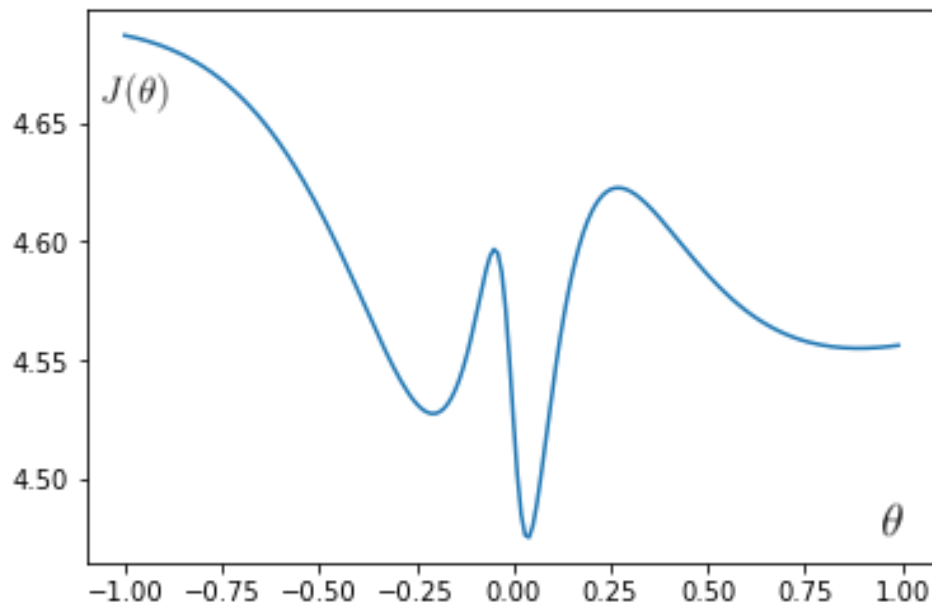
Linear regression:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} \left( h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$
$$= \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$


$$\text{Cost}(h_{\theta}(x), y) = \frac{1}{2} (h_{\theta}(x) - y)^2$$

## Cost function

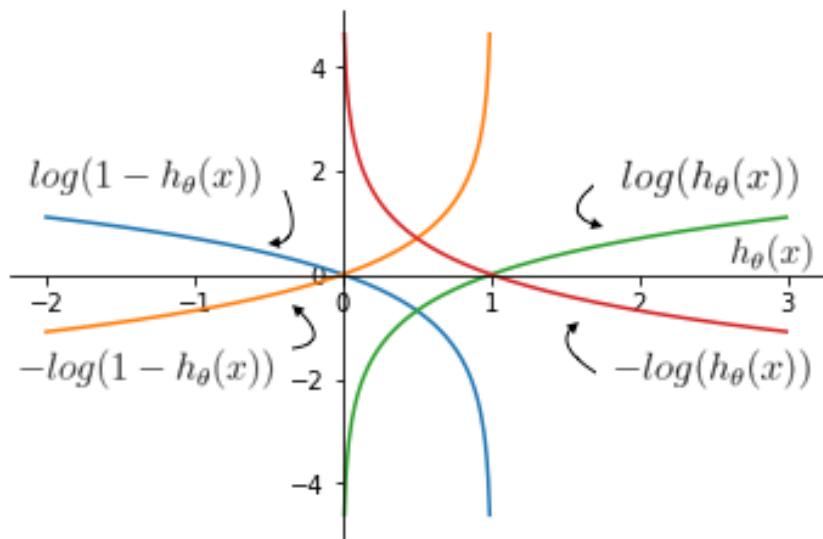
Linear regression: 
$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$



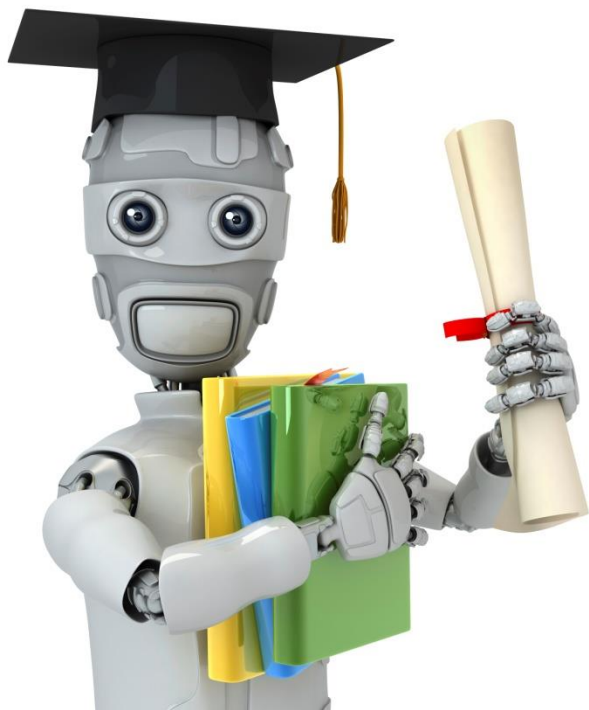
**Non convex cost**

# Logistic regression cost function

$$Cost(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



- If  $y = 1$  and
  - $h_{\theta}(x) = 1$ , then  $Cost = 0$
  - $h_{\theta}(x) \rightarrow 0$ , then  $Cost \rightarrow \infty$
- If  $y = 0$  and
  - $h_{\theta}(x) = 0$ , then  $Cost = 0$
  - $h_{\theta}(x) \rightarrow 1$ , then  $Cost \rightarrow \infty$



Machine Learning

# Logistic Regression

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Simplified cost function  
and gradient descent

## Logistic regression cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

Note:  $y = 0$  or  $1$  always


$$\text{Cost}(h_{\theta}(x), y) = -y \log(h_{\theta}(x)) - (1 - y) \log(1 - h_{\theta}(x))$$



## Logistic regression cost function

$$\begin{aligned} J(\theta) &= \frac{1}{m} \sum_{i=1}^m \frac{1}{2} \left( h_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \\ &= \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \\ &= -\frac{1}{m} \sum_{i=1}^m \left( y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right) \end{aligned}$$

## Logistic regression cost function

$$\begin{aligned} J(\theta) &= \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \\ &= -\frac{1}{m} \left[ \sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right] \end{aligned}$$

To fit parameters  $\theta$ :

$$\min_{\theta} J(\theta)$$

To make a prediction given new  $x$ :

$$\text{Output } h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

# Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[ \sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

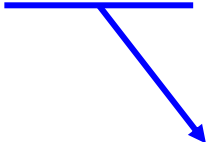
Want  $\min_{\theta} J(\theta)$ :

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

(simultaneously update all  $\theta_j$ )


$$\frac{1}{m} \sum_{i=1}^m \left( h(x^{(i)}) - y^{(i)} \right) x_j^{(i)}$$

$$\begin{aligned}
\frac{\partial}{\partial \theta} J(\theta) &= -\frac{\partial}{\partial \theta} \frac{1}{m} \sum_{i=1}^m \left( y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right) \\
&= -\frac{1}{m} \sum_{i=1}^m \left( y^{(i)} \frac{\partial}{\partial \theta} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \frac{\partial}{\partial \theta} \log(1 - h_{\theta}(x^{(i)})) \right)
\end{aligned}$$

$$\frac{d}{dz} \log(z) = \frac{1}{z}$$

$$\begin{aligned}
\frac{d}{dz} h(z) &= \frac{d}{dz} \frac{1}{1 + e^{-z}} \\
&= \frac{e^{-z}}{(1 + e^{-z})^2} \\
&= \frac{1 + e^{-z} - 1}{(1 + e^{-z})^2} \\
&= \frac{1}{1 + e^{-z}} - \frac{1}{(1 + e^{-z})^2} \\
&= h(z)(1 - h(z))
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial \theta_j} J(\theta) &= -\frac{1}{m} \sum_{i=1}^m y^{(i)} \frac{x_j^{(i)}}{h(x^{(i)})} h(x^{(i)}) (1 - h(x^{(i)})) + (1 - y^{(i)}) \frac{x_j^{(i)}}{1 - h(x^{(i)})} (-h(x^{(i)}) (1 - h(x^{(i)}))) \\
&= -\frac{1}{m} \sum_{i=1}^m (y^{(i)} (1 - h(x^{(i)})) - (1 - y^{(i)}) h(x^{(i)})) x_j^{(i)} \\
&= -\frac{1}{m} \sum_{i=1}^m \left( y^{(i)} - y^{(i)} h(x^{(i)}) - h(x^{(i)}) + y^{(i)} h(x^{(i)}) \right) x_j^{(i)} \\
&= \frac{1}{m} \sum_{i=1}^m \left( h(x^{(i)}) - y^{(i)} \right) x_j^{(i)}
\end{aligned}$$

## Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[ \sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want  $\min_{\theta} J(\theta)$ :

Repeat {

$$\theta_j := \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

} (simultaneously update all  $\theta_j$ )

Algorithm looks identical to linear regression!

# Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[ \sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want  $\min_{\theta} J(\theta)$ :

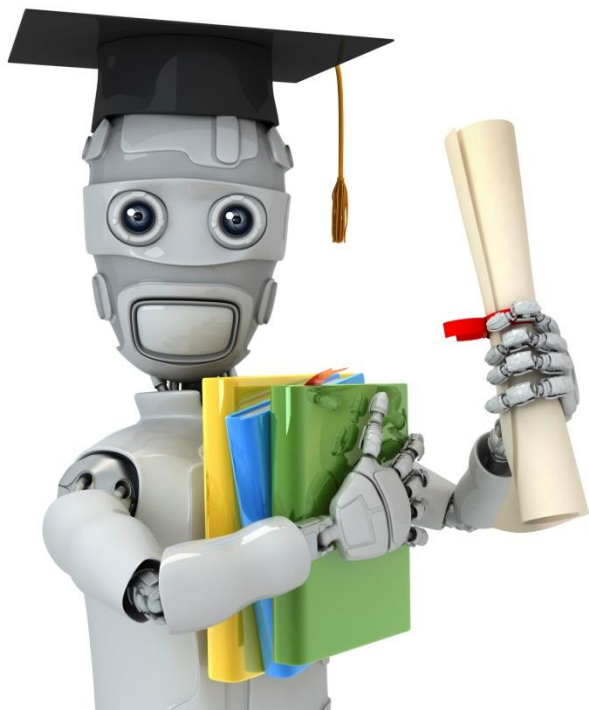
Repeat {

$$\theta_j := \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

}

vectorized implementation

$$\theta := \theta - \alpha \frac{1}{m} X^T (g(X\theta) - y)$$



Machine Learning

# Logistic Regression

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Multi-class classification:  
One-vs-all



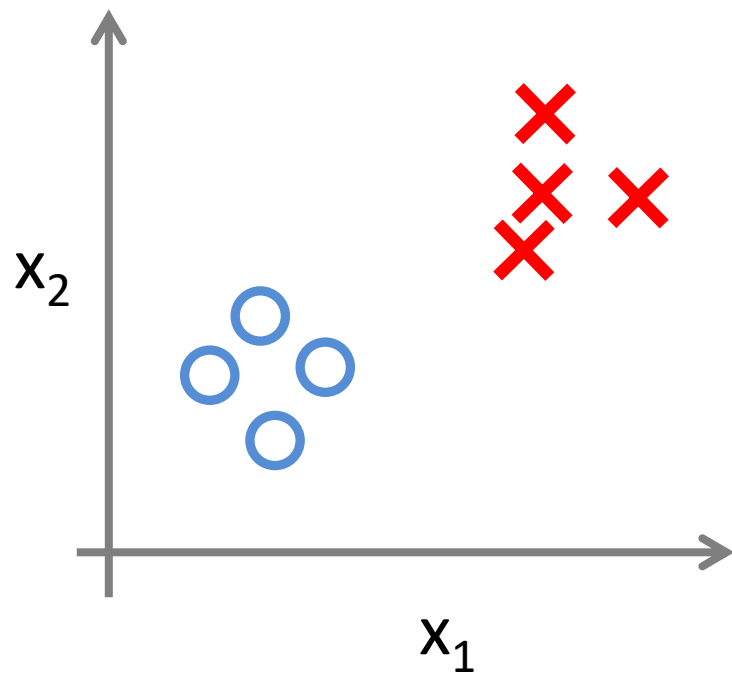
## **Multiclass classification**

Email foldering/tagging: Work, Friends, Family, Hobby

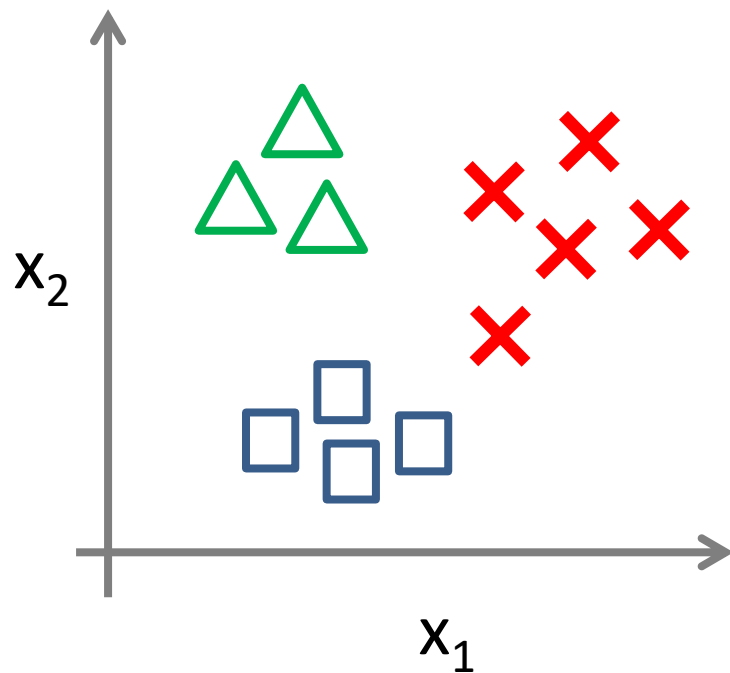
Medical diagrams: Not ill, Cold, Flu

Weather: Sunny, Cloudy, Rain, Snow

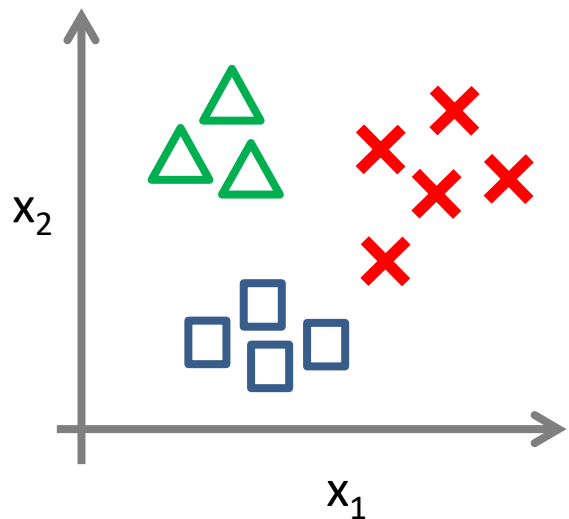
Binary classification:



Multi-class classification:



## One-vs-all (one-vs-rest):

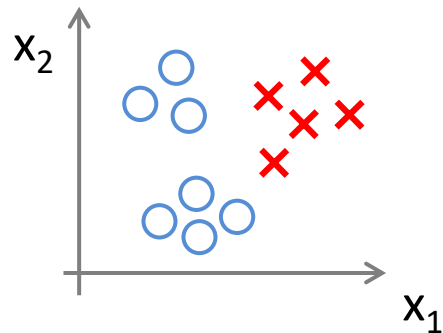
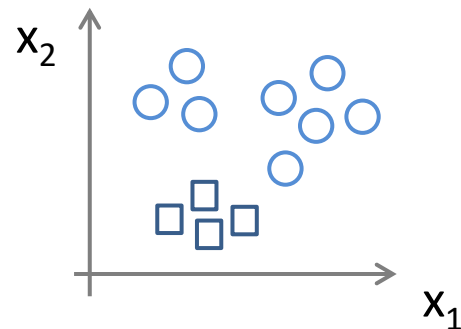
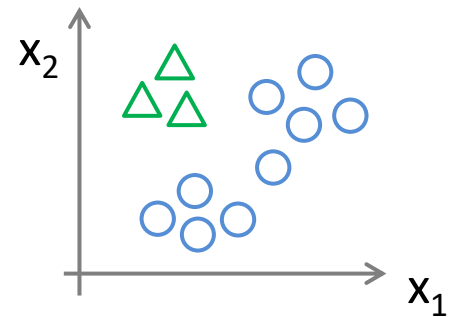


Class 1: 

Class 2: 

Class 3: 

$$h_{\theta}^{(i)}(x) = P(y = i|x; \theta) \quad (i = 1, 2, 3)$$



## One-vs-all

Train a logistic regression classifier  $h_{\theta}^{(i)}(x)$  for each class  $i$  to predict the probability that  $y = i$ .

On a new input  $x$ , to make a prediction, pick the class  $i$  that maximizes

$$\max_i h_{\theta}^{(i)}(x)$$